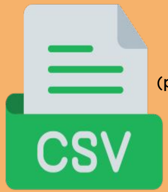
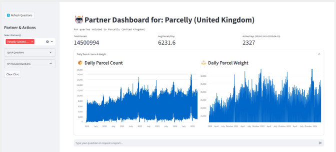
System Pipeline – End-to-End Overview



(parcel_per_network_partner.csv)

ETL Layer (etl.py)
1. Clean & Transform
2. Normalize names
3. Persist to DuckDB

Streamlit UI Layer (main.py)
1. Partner selection
2. Metrics and Charts



LLM Agent Layer (agent.py)

1. Langchain + Deepseek-chat

2. Dynamic Key performance indicator (KPI) prompting

3. Single vs Multi-partner

Chat interface + visual response

1. Simple markdown based reponses

2. Trends, summaries, table evaluation

3. User feedback integration

Key Assumptions

📊 **Data Consistency**
Assumed all CSV files follow a fixed schema (date, item, weight, partner) without needing schema validation.

🗣️ **English-Only Interface**
Assumed all users can interact fluently in English—no localization or translation was implemented.

🕒 **Static Historical Analysis**
Assumed historical data was sufficient; real-time updates or forecasting were not required for the MVP.

📦 **Fixed Partner List**
Assumed the partner names are known and stable—handled via hardcoded alias mappings.

🤖 **LLM Stability**
Assumed DeepSeek's API would remain consistently available and responsive throughout the pilot.

MVP design decisions and trade-offs

Design Decision	Trade-off	Justification
Streamlit for UI	Limited UI control	Fast prototyping and native pandas/chart support
CSV + DuckDB for data	No real-time sync	Lightweight and easy to manage in MVP
DeepSeek LLM via API	3rd-party dependency	Easy integration, cost-effective experimentation
No authentication	No access control	Internal-use focus for initial pilot
Hardcoded alias mapping	Scalability limitations	Simplified matching in MVP
English-only language support	Regional language barriers	Reduced development effort for early release
Descriptive analytics only	No predictions or future modeling	Simpler, interpretable responses for users

Possible improvements in future

🔍 **Retrieval-Augmented Generation (RAG)**

- Combine LLM model with DuckDB stores for source-aware responses.

📈 **Forecasting Model Support**

- Integrate time-series forecasting (Prophet, LSTM) for parcel prediction in partner locations.

🌍 **Multilingual Chatbot**

- Support French, German, Italian using translation APIs or multilingual LLMs (agentic framework maybe?).

🧾 **Scheduled Reports & PDF Exports**

- Enable downloadable insights and automatic report generation for partners and stake-holders.

🔐 **Authentication**

- Add login and role-based dashboard access for secure use.